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$$\int_{0}^{+\infty} e^{-ax}dx \left(a \in \mathbb{R}_0^+\right)$$

$$= \lim_{b \to +\infty} \int_{0}^{b} e^{-ax}dx$$

$$u = -ax \Rightarrow du = -adx \Rightarrow dx = -\frac{du}{a}$$

$$= \lim_{b \to +\infty} -\frac{1}{a} \int_{0}^{b} e^{u}du$$

$$= \lim_{b \to +\infty} -\frac{1}{a} \left[e^{u}\right]_0^b$$

$$= \lim_{b \to +\infty} -\frac{1}{a} \left[e^{-ax}\right]_0^b$$

$$= \lim_{b \to +\infty} -\frac{1}{a} \left[e^{-ab} - 1\right]$$

$$= -\frac{1}{a} \left[0 - 1\right]$$

$$= \frac{1}{a}$$

References